

Guangyu (Daniel) Hou

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EDUCATION

New York University New York, USA	Sep. 2025 –
<i>Master of Science in Mathematics in Finance, NYU's Courant Institute of Mathematical Sciences</i>	
Nankai University Tianjin, China	Sep. 2021 – Jun.2025
<i>B.S. in Mathematics and Applied Mathematics, GPA: 3.61/4.0, Ranking: 23/92</i>	
Scholarships & Awards:	
• Scholarship of Social Welfare, Nankai University	Dec. 2023
• Outstanding Student Cadre, Nankai University Student Union	Sep. 2023
• National Undergraduate Innovation and Entrepreneurship Project Award	Jan. 2023
• Academic Excellence Award, Shiing-Shen Chern Class, School of Mathematical Sciences, NKU	Sep. 2021 – Sep. 2022
Relevant Courses: <i>Real Analysis(91), ODE(92), Python(94), Operations Research (91), Financial Options(96), Stochastic Operations Research(90), Game Theory(100), PDE(97)</i>	
University of California, Berkeley CA, USA	Jan. 2024 – Jul.2024
BERKELEY GLOBAL ACCESS (BGA) Program, GPA: 3.88/4.0	
Relevant Courses: <i>Stochastic Processes(A), Graduate level: Probability Theory(A-), Non-stationary Dynamics(A)</i>	

RESEARCH AND PROJECT EXPERIENCE

Feasibility of Transfer Learning and Model Adaptability	Apr. 2024 – Oct.2024
<i>Research Assistant, Advised by Prof. Haoyang Cao from Johns Hopkins University</i>	
• Explored Optimal Transport to address domain adaptation challenges, which form the foundation of transfer learning • Studied Mean Field Equilibrium to efficiently approximate n-player Nash equilibria, with a particular emphasis on their benefits and underlying procedures • Investigated applications of Mean Field Games across various models, analyzing how to design specific types of games and their potential real-world implementations	
New Media Marketing Strategy for Time-Honored Chinese Brands	Jan. 2023 – Mar.2025
<i>Team Member, China National University Student Innovation & Entrepreneurship Development Program</i>	
• Conducted consumer sentiment analysis to explore methods for understanding the emotions behind consumer reviews, providing strategic direction for new media marketing • Utilized Python for data collection and analysis, focusing on key performance indicators to evaluate the current state of new media marketing for time-honored Chinese brands • Performed field studies to identify the most cost-effective and impactful new media marketing methods for these brands, comparing them with industry benchmarks	
Apple Image Recognition and Quality Assessment	Nov. 2023 – Dec.2023
<i>Course Project, Nankai University</i>	
• Used Spectral Clustering in Python to treat data points as a weighted graph, maximizing the edge weights within the same subgraph and minimizing the edge weights between different subgraphs • Defined ripeness and quality based on surface blemishes, size, and HSV color attributes (Hue, Saturation, Value) for precise maturity clustering	

INTERNSHIP EXPERIENCE

Fintech Intern, ZHONGTAI SECURITIES

Jun. 2025 – Present

Fintech Research institute of Zhongtai Securities, Supervisor: Ms. Qianyun Li

- Constructed backtesting framework for fund evaluation factors, developed quantitative strategies
- Assisted in building foundational models on macroeconomic and capital market data

Strategy & Operations Analyst Intern, DiDi

Mar. 2025 – May.2025

Two-Wheel Vehicle Division, Supervisor: Ms. Lili Gao

- Monitored the operational status and vehicle distribution of shared bicycles in Tianjin by analyzing dashboard data
- Strengthened Excel proficiency
- Configured screening metrics designated by headquarters, adjusted weightings accordingly, and collaborated with the operations team to maximize overall efficiency

Fintech Intern, China Everbright Bank

Sep. 2024 – Nov.2024

Data Asset Management Department, Supervisor: Mr. Zifan Wang

- Studied reinforcement learning theories and algorithms systematically
- Analyzed and derived theoretical implications of algorithmic evolution, assessing their potential for large language model (LLM) integrations
- Experimented with combining the Actor- Critic algorithm with the PPO algorithm to enhance the performance of open-source large language models within the banking industry

Data Analyst and Quant Developer Intern, China Fund Management Co., Ltd

Jul. 2024 – Sep. 2024

Index and Quantitative Investment Department, Supervisor: Ms. Tingting Ren

- Developed Python scripts and optimized PySpark workflows to filter out anomalies and missing data in stock futures from the Wind data source
- Assessed data accuracy and reliability from Ricequant platform
- Integrated the Tonghuashun API to replicate Wind API functionalities, enabling strategy execution, order placement, and cross-validation of data for enhanced system reliability

EXTRACURRICULAR ACTIVITY

Secretariat staff, Berkeley Chinese Students and Scholars Association (BCSSA)

Jan. 2024 – May. 2024

- Organized and planned the online concert “Eclipse,” achieving over a thousand views through effective promotion and real-time attendance management

Vice Minister of Sports Department, Student Union of Nankai University

Oct. 2021 – Jun.2023

- Led and coordinated major sports events at Nankai University
- Managed promotion, resource preparation, personnel coordination and training, and on-site scheduling, successfully attracting over a thousand students to participate in events each year
- Honored as the “Outstanding Student Cadre” of the union

SKILLS AND HOBBIES

Technical Skills: Excel, Python, C++, LaTeX, Xiumi (for professional document formatting)

Language: Mandarin(Native), English(TOEFL 112: R30, L28, S25, W29; CET-6 623)

Hobbies: Basketball, Badminton, Vamp